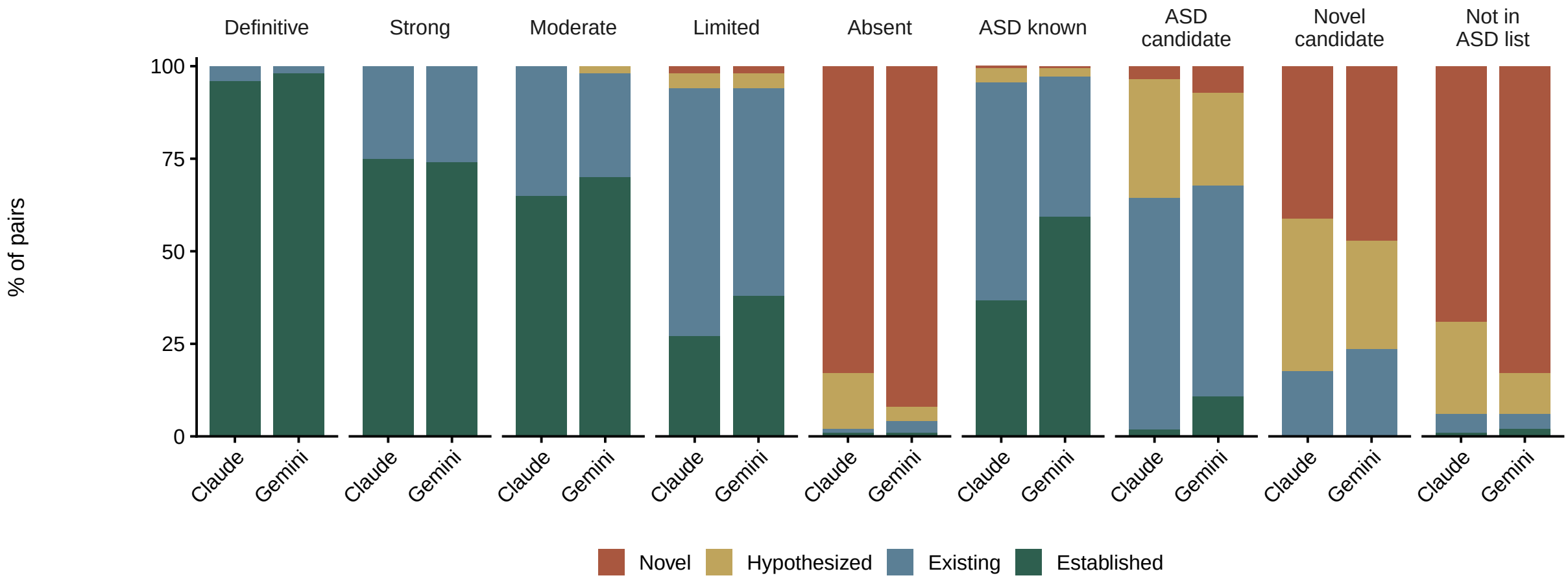
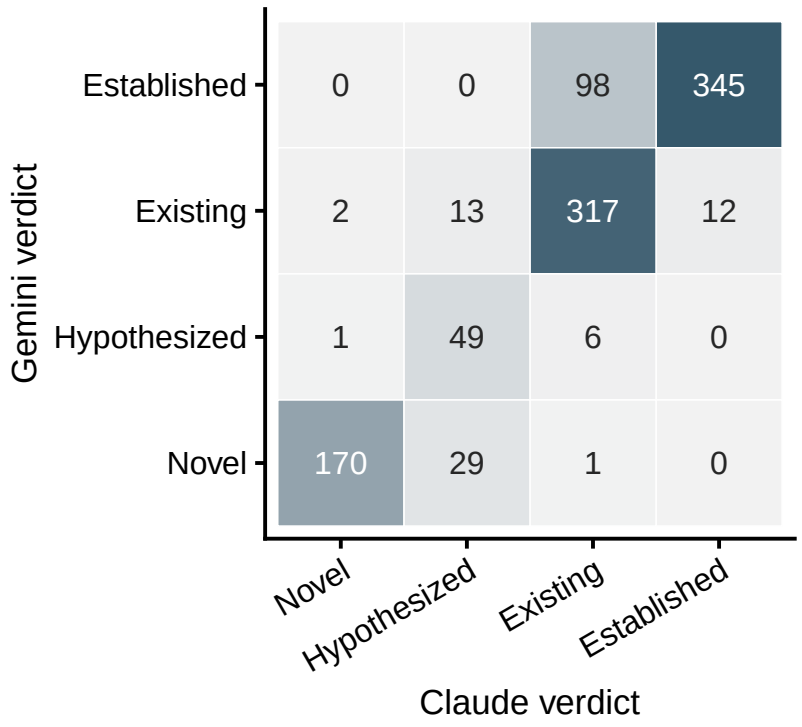
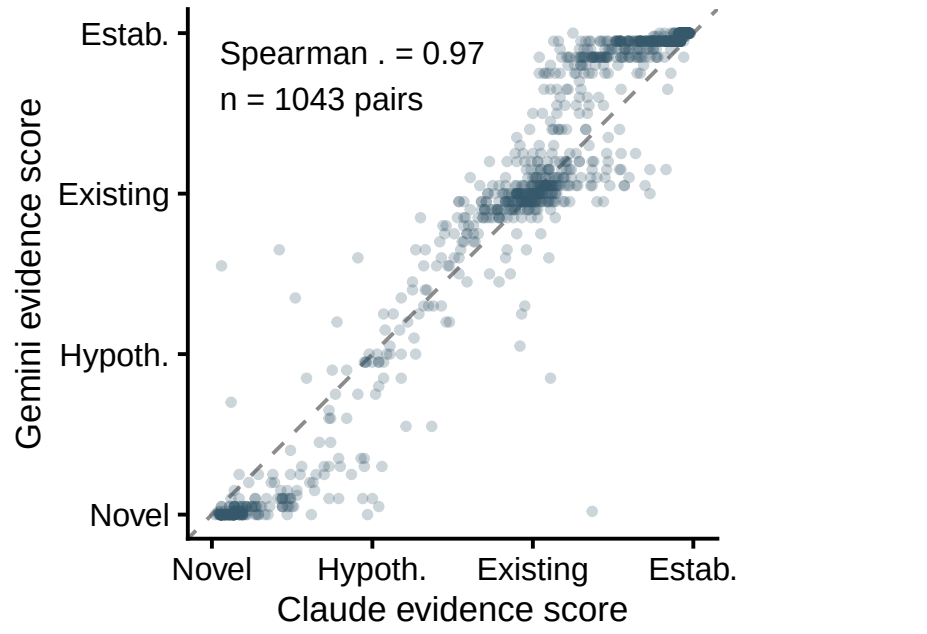
a


Definitive
Strong
Moderate
Limited
Absent
ASD known
ASD candidate
Novel candidate
Not in ASD list
% of pairs
100
75
50
25
0
Claude
Gemini
Novel
Hypothesized
Existing
Established

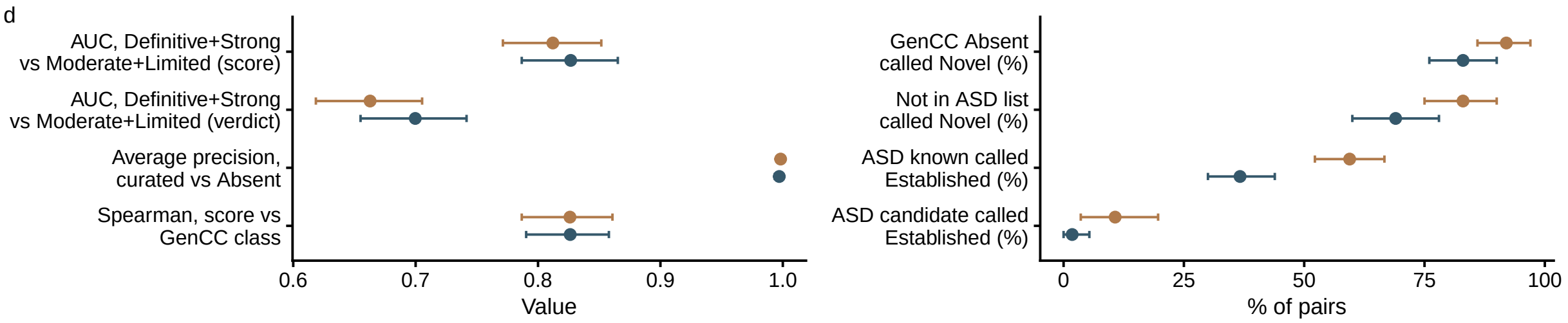
b


Established
Existing
Hypothesized
Novel
Gemini verdict
0
0
98
345
2
13
317
12
1
49
6
0
170
29
1
0
Novel
Hypothesized
Existing
Established
Claude verdict

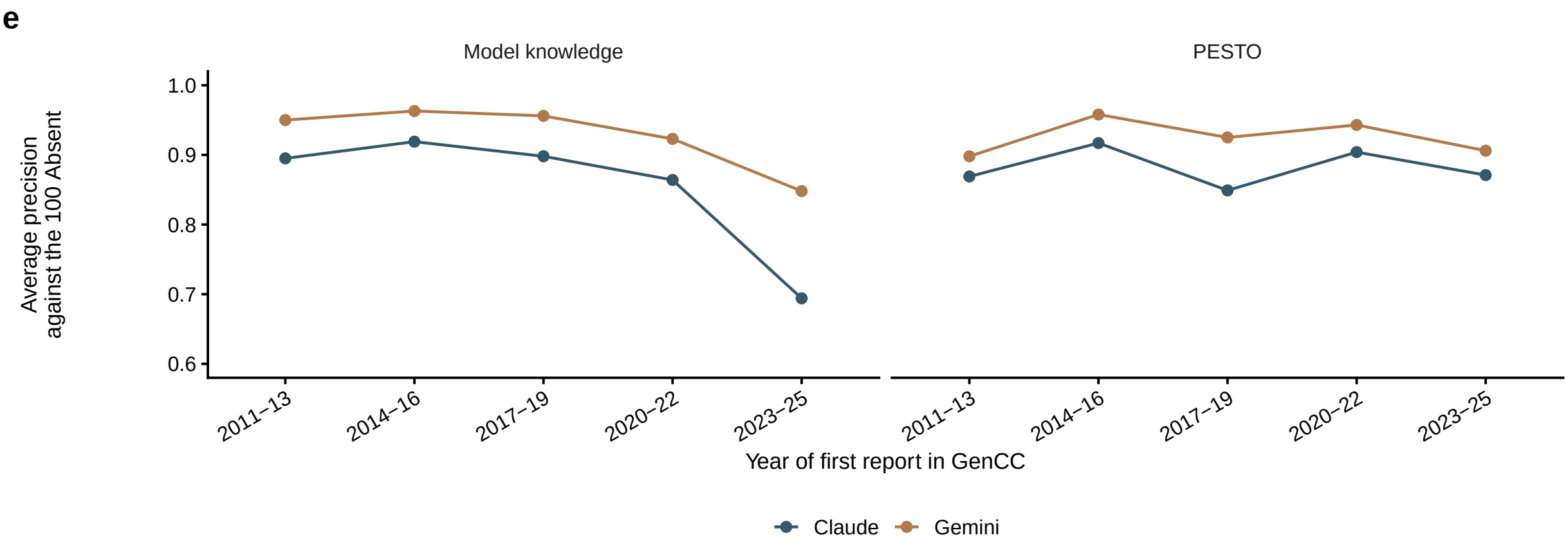
c


Estab.
Existing
Hypoth.
Novel
Gemini evidence score
Spearman . = 0.97
n = 1043 pairs
Novel
Hypoth.
Existing
Estab.
Claude evidence score

d


AUC, Definitive+Strong vs Moderate+Limited (score)
AUC, Definitive+Strong vs Moderate+Limited (verdict)
Average precision, curated vs Absent
Spearman, score vs GenCC class
0.6
0.7
0.8
0.9
1.0
Value
GenCC Absent called Novel (%)
Not in ASD list called Novel (%)
ASD known called Established (%)
ASD candidate called Established (%)
0
25
50
75
100
% of pairs

e


Model knowledge
PESTO
Average precision against the 100 Absent
1.0
0.9
0.8
0.7
0.6
2011–13
2014–16
2017–19
2020–22
2023–25
Year of first report in GenCC
Claude
Gemini